



The recognition of read and spontaneous speech in local vernacular: The case of Zurich German



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ABSTRACT

Listeners are typically able to identify speech as either being produced spontaneously or read from a transcript. In the present research we investigated whether this is true in vernacular speech when typical cues to read and spontaneous speech are either missing and/or ambiguous. In addition it was investigated what the acoustic cues for listeners' identification ability were. 26 listeners of Zurich German participated in two perception experiments. In Experiment I, listeners judged 128 stimuli (64 spontaneous, 64 read) in a two-alternative identification task as either spontaneous or read. Results revealed that overall listener performance was well above chance (mean $A' = 0.82$) while there was no bias for either read or spontaneous speech (mean $B_D = 0$). There were significant effects of speaker and listener in A' and B_D . In Experiment II, the same 26 listeners rated the same 64 read speech stimuli from Experiment I as to whether they sounded more or less read. Results revealed that there was considerable within-category variability as a function of speaker. From eight acoustic prosodic parameters only articulation rate explained listener behavior to some degree in Experiments I and II. Overall the study suggested that read and spontaneous speech can be recognized based on very subtle cues to these speaking styles.

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1. Introduction

The present study was directly motivated from our previous research on speaker-individual speech rhythmic characteristics in which we investigated the variability of speech rhythm within and between speakers based on the *Temporal Voice Idiosyncrasy Corpus* (henceforth: TEVOID Corpus; cf. Dellwo, Leemann, & Kolly 2012; Leemann, Kolly, & Dellwo 2014). In this speech corpus we created acoustic within-speaker variability by asking speakers to produce spontaneous and read speech in Zurich German, two production modes of speech that are generally referred to as *speaking styles* in the literature (Listerri, 1992; Laan, 1997). Both tasks were performed in our laboratory. We first recorded speakers in an interview situation. From these recordings we extracted a number of sentences that were free from grammatical errors, semantically coherent and fluently produced, and created a written transcript for these sentences. In a second recording session (a few months later), each speaker read all sentences from the transcript (his/her own sentences as well as those previously produced by his/her peers). We carried out a large variety of durational measurements that are typically referred to as rhythm measures (Ramus, Nespor, & Mehler, 1999; Grabe & Low, 2002). Results from measurements of these variables consistently showed strong between-speaker but little or inconsistent within-speaker effects between spontaneous and read speech (Dellwo et al., 2012; Leemann et al., 2014). We thus argued that the rhythmic durational characteristics we observed do not vary between read and spontaneous speech. There is, however, an alternative interpretation possible: the speech produced under the strict laboratory settings might be poor examples of the categories *read* and *spontaneous* and might thus contain poor cues to these speaking styles. In the present research we followed up this interpretation and hypothesized that if this was the case then listeners should show a poor performance in identifying the speech from the TEVOID corpus as either read or spontaneous.

What are the reasons to assume that the speech elicited under the conditions described above might not contain sufficient cues to speaking style? It has often been argued that read and spontaneous speech are not members of two distinct categories but that there

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is a continuum reaching from canonical spontaneous speech, e.g. a casual conversation in the street, to canonical read speech, e.g. the speech of a news reader (see review in [Laan, 1997](#)). In terms of production, numerous mechanisms may lead to strong variability within the read and spontaneous categories. It might be, for example, that speakers take on different styles in their reading (news style, theatrical style, etc.) or read speech might also vary depending on who is addressed with it ([Niebuhr et al., 2010](#)). The same seems plausible for spontaneous speech. In addition, the labels “read” and “spontaneous” are not well defined categories but are often subcategorized. [Labov \(1972\)](#), for example, distinguishes, amongst others, between reading style, careful speech, casual speech and spontaneous speech. [Smiljanic and Bradlow \(2009\)](#) further distinguish between casual and clear speech, which, in laboratory settings, are typically both variants of read speech. Clear speech is intended to be particularly intelligible to listeners with listening impairments or other speech comprehension problems. Even though the labels “spontaneous” and “read” might seem applicable for the speech in our TEVOID corpus – we know it for a fact that read speech was read and spontaneous speech was produced spontaneously – it seems that our speaking styles are not very far apart on a possible read-spontaneous continuum. We argue that this is mainly the result of (a) a lack of acoustic cues peculiar to read/spontaneous speech due to our data-collection method and (b) ambiguous acoustic cues to read and spontaneous speech due to the use of vernacular speech. In the following we give support for both reasons:

Re (a): Our spontaneous speech was recorded in a rather formal recording situation (recording booth, semi-structured interview) and our sentence selection criteria of grammatical correctness, and semantic coherence and fluency (see above) may have removed important cues to this style such as false starts, rhythmic irregularity, phrase-final voicing/devoicing or timing phenomena (see discussion in [Laan, 1997: 44](#)). Such a drastic reduction of cues to spontaneity is typically not present in previous studies. In [Mixdorff and Pfitzinger \(2005\)](#) listeners identified speech as either read or spontaneous. Like in our method Mixdorff and Pfitzinger asked listeners to read scripted utterances they previously produced spontaneously. However, their spontaneous speech contained characteristics such as filled and unfilled pauses and they reported that in particular filled pauses such as [E:m] facilitated its identification (p. 317). This means that the most important cues to spontaneous speech in Mixdorff and Pfitzinger's study were not present in our data. The same is true for numerous other cues reported in the literature: [Cucchiari, Strik, and Bou \(2002\)](#) used mainly different characteristics of silent pauses and hesitation markers in quantifying fluency cues in read and spontaneous speech. [Liu and Zeng \(2006\)](#) found that the insertion of silent gaps into conversational speech increases its intelligibility and thus makes it more appear like clear speech. [Hirschberg \(2000\)](#) found that about 20% of English spontaneous sentences were characterized by disfluencies which might be used for their identification. [Pettinato and Hazan \(2013\)](#) and [Hazan and Pettinato \(2014\)](#) studied the use of pausing behavior in speakers' attempts to make their speech sound more clear and found that it was used both by adults and children (even though to different degrees). Finally, [Levin, Schaffer, and Snow \(1982\)](#) found that listeners could identify read and spontaneous speech with an accuracy of 84% mainly based on pause characteristics, false starts, wording and grammar. Given the lack of such prominent indicators in our non-disfluent spontaneous speech it is unclear whether listeners are still able to distinguish it from read speech.

Re (b): Another peculiarity in our data lies in the ambiguity of cues to spontaneity/reading resulting from the use of vernacular speech. We asked Swiss German speakers from Zurich to read in their local vernacular of Zurich German¹. In Swiss German two forms of the same language – a high variety and a low variety – co-exist. This situation has been described as one of “medial diglossia” ([Kolde, 1981](#)): Standard German is mainly used in written form while Swiss German is used for verbal communication. Swiss German, however, is viewed as more prestigious than Standard German ([Sieber & Sitta, 1986](#)) even though it does not feature a formally-defined spelling system. Writing in Swiss German is largely restricted to recent technological inventions, including e-mailing, mobile text messaging, social media, and chat-rooms ([Christen, 1999](#); [Siebenhaar, 2003](#)). Younger speakers are therefore used to some degree to reading texts in vernacular but typically in situations that are very related to settings in which spontaneous speech would occur. Swiss Germans typically acquire the vernacular variety first in life – and this was the case with all our speakers – the standard variety of German is later acquired at school. To some degree speakers become acquainted to it prior to school through the media. So Standard German has strong similarities with an L2 for Swiss speakers ([Häcki Buhofer & Burger, 1998](#)). Given this situation, Swiss German speakers might be particularly prone to read texts written in Swiss German in a spontaneous way. In addition, and possibly more importantly, when Swiss German listeners come across vernacular speech they might be strongly biased towards identifying this speech as spontaneous. This might be additionally evoked by lexical markers to spontaneous speech such as the adverb *dänn* (closest English translation: ‘then’) in the sentence *Ich bin wäge Sprachwüßeschaft dänn usegheit* (English: ‘I then failed because of linguistics’) which is not typically used in formal writing, or the modal particle *ja* (closest English translation: ‘as you know’) in *Chasch ja nöd nöime andersch go studiere mit Erasmus* (‘You cannot go studying somewhere else with Erasmus’). It might also have been evoked by syntactical constructions that are typically not found in written language. Further, we might have introduced ambiguity in style cues through our reading instructions. [Cucchiari et al. \(2002\)](#) reported strong differences between the fluency of speakers reading speech without prior rehearsal. In our case all speakers practiced reading the sentences prior to the recordings to be able to read them fluently. This means that in particular temporal variability present in

¹ Even though in Swiss German the vernacular varieties are generally referred to as ‘dialects’ they reveal more strongly the characteristics of a vernacular as they particularly lack the typical continuous variability between dialectal variety and standard.

more disfluent read speech was strongly reduced in our data. In return, it might have caused speakers to produce the vernacular sentences in a more typical vernacular mode which should be closer to spontaneous than to read speech.

In summary, the data in our TEVOID corpus contains spontaneous speech that lacks important cues to spontaneity (pauses and fillers) and read speech in vernacular that is typically not read and contains lexical and syntactic cues to spontaneous speech. The speech thus lies on the borderline between the two speaking styles. In Experiment I (Section 3) we thus tested how reliably listeners of Zurich German identified the speech as either read or spontaneous. We also wanted to know whether the spontaneous nature of the read speech might create a bias in listeners towards spontaneous speech when listening to read speech from their vernacular. Finally, we were interested in the degree to which the variability can be explained by the performance of different speakers and listeners and what the acoustic cues to read and spontaneous speech in our data were?

The lack and the ambiguity of cues to read and spontaneous speech in our data must inevitably lead to weak between-category and considerable within-category variability. This must be particularly true for the read speech as it was of spontaneous nature (vernacular speech) and it contained lexical and syntactic cues to spontaneous speech. Even though speakers use their vernacular in informal writing situations and even though we took measures to ensure that speakers can read our experimental sentences fluently, there may be subtle fluency differences, which could be an audible marker for the read quality of their speech (Cucchiari *et al.*, 2002; Wolf & Katzir-Cohen, 2001). We therefore investigated within-category variability in read speech in Experiment II (Section 4) where listeners rated the read speech by speakers from Experiment I on a continuous scale that reached from “sounds rather less read” to “sounds rather more read”. We were particularly interested in whether the percept of “more or less read” could be explained by acoustic variability between the stimuli or whether there is reason to believe that particular speakers reveal characteristics in their reading that makes them appear as more or less read (or both). In Experiment II we also focused on another factor that may contribute to within-category variability of read speech. In our method, speakers read sentences which they previously either uttered spontaneously themselves (henceforth: same origin) or which were uttered spontaneously by other speakers (henceforth: different origin). It seems conceivable that sentences which originated from the same speaker contained grammatical structures, lexical choices and semantic characteristics that the reader is familiar with and thus might find easier to re-produce fluently. Finally, we wanted to know whether there is a relationship between the accurateness in the identification of a stimulus as being read and its perceived readness rating. We hypothesized that stimuli that are more accurately identified as read speech should also be the ones that are rated as most strongly read.

2. Acoustic cues to read and spontaneous speech

Given that the segmental inventory was nearly identical between the styles because of the re-reading task, it seems that prosodic characteristics in particular should differ between the styles. Spontaneous speech is optimized for communication and shows in particular greater suprasegmental variability than read speech (Cucchiari *et al.*, 2002; Lieberman, Katz, Jongman, Zimmerman, & Miller, 1985; Shriberg, 2005). Levin *et al.* (1982) showed that the frequency-temporal characteristics of F0 contours in low-pass filtered speech contained enough acoustic cues for listeners to distinguish between read and spontaneous speech. Remez, Rubin, and Ball (1985), and Remez, Rubin, and Nygaard (1986) also used transcripts of spontaneous sentences for eliciting read speech. Even though no detailed results have been reported, the authors state that the identification ability of listeners was generally high. This was true even when stimuli were resynthesised with a monotonous pitch, which would give rise to the assumption that read and spontaneous speech can be identified even when only rudimentary prosodic cues are present. Cucchiari *et al.* (2002) reported that especially temporal cues like rate and pausing behavior are responsible for the fluency of read and spontaneous speech. Such findings motivated us to analyze prosodic variables in our stimuli in particular. While pausing behavior was hardly available in our data we focused our attention on rate in particular. A variety of prosodic variables have been reported as relevant in the detection of clear speech (Liu, Del Rio, Bradlow, & Zeng, 2004; Payton & Braida, 1999; Payton, Uchanski, & Braida, 1994) and also in the read/spontaneous speech distinction (Laan, 1997). From these studies we comprised the following set of variables:

- *Articulation rate*: Numerous studies reported variability of articulation rate between read and spontaneous speech (Hirschberg, 2000; Laan, 1997; Mixdorff & Pfitzinger, 2005) as well as clear and conversational speech (Hazan & Pettinato, 2014; Pettinato & Hazan, 2013; Smiljanic & Bradlow, 2009). In the present paper we took a traditional and widely applied approach in measuring articulation rate as the number of syllables per second.
- *Fundamental frequency characteristics*: The mean absolute differences and the variability of fundamental frequency has been reported to be an indicator of read and spontaneous speech (Laan, 1997). Hirschberg (2000) reported fundamental frequency contours (in ToBI annotation, Silverman *et al.*, 1992) to vary between utterances. Declarative sentences showed, for example, more LL% contours in read speech compared to spontaneous and so did wh-questions in one of the corpora under investigation. However, a large overlap between the categories could be observed. Since in our stimulus set (128 stimuli, see Section 3.1) sentence type did not vary sufficiently we did not find this method applicable. We thus decided to apply approaches like that in Laan (1997) and measured the mean fundamental frequency and its variability.
- *Intensity characteristics*: There is strong evidence in the literature that intensity varies between read and spontaneous speech. Mostly this variability is measured by the variability in the modulation index of the intensity envelope for a variety of different modulation frequencies (Houtgast & Steeneken, 1985; Liu *et al.*, 2004; Payton *et al.*, 1994). Payton *et al.* found, for example, that

clear speech reveals stronger temporal amplitude modulations than conversational speech. However, such variability is strongly correlated with articulation rate. We applied a simple version of the modulation index by calculating the intensity variability for the entire frequency spectrum with two different variables: Instead of a continuous modulation, we calculated intensity mean values for the vocalic sections in the signal and obtained a coefficient for the intensity variability between these sections. To monitor the distribution of the energy across different frequency bands we measured the slope of the long-term average spectrum LTAS. A steeper slope indicates less energy at higher frequency regions, a flat slope indicates higher energy.

In addition to the measures above we also investigated suprasegmental timing characteristics (speech rhythm). In our previous work we found strong and consistent variability of these variables between speakers, however, there was only rudimentary variability between read and spontaneous speech (Dellwo et al., 2012; Leemann et al., 2014). For this reason we did not expect that listeners can make use of these characteristics. To test whether this finding can be replicated for the stimulus set of the present study, we included three variables that have been widely applied in the literature on speech rhythm and have typically been reported to vary strongly between languages (Dellwo, 2010; Grabe & Low, 2002; Ramus et al., 1999):

- *Vocalic-consonantal ratio*: The ratio of the overall time spent on vowels compared to consonants (%V, Ramus et al., 1999).
- *Consonantal durational variability*: Consonantal durational variability was measured by the standard deviation of consonantal intervals (deltaC, Ramus et al., 1999).
- *Vocalic durational variability*: Vocalic durational variability was measured by the rate normalized Pairwise Variability Index (nPVI_V; Grabe & Low, 2002).

3. Experiment I

In the present experiment we tested whether and to what degree listeners were able to identify read and spontaneous speech from the TEVOID Corpus. We further examined the variation in performance as a function of speaker and as a function of listener. To be able to distinguish between listeners' sensitivity and bias we applied Signal Detection Theory measures (Green & Swets, 1966). As a correlate for sensitivity we used A' and as a correlate of listeners' bias we measured B_D' (Pallier, 2002).

By now there is a wide body of evidence in the literature showing that read and spontaneous speech can be distinguished by listeners (Levin et al., 1982; Mixdorff & Pfitzinger, 2005). Like in our study, Laan (1997) worked with read transcripts of spontaneous speech. In a screening experiment carried out to find speech that is categorized best as read and spontaneous, she showed that for two speakers high identification scores (67.7% and higher) were achieved. Laan's experiment, however, contained the read and the spontaneous version of literally identical utterances (even though they were separated in the experimental presentation by at least three stimuli). This means that listeners were to some degree able to contrast both versions (read and spontaneous) of the same sentence. While this method might have been very applicable for the selection of suitable sentences for the acoustic manipulation, we found that, in our experiment, listeners should not hear the read and spontaneous version of the same sentence. This requires listeners to use abstract knowledge about read and spontaneous speech that they apply to the decision process of each individual stimulus.

3.1. Method

3.1.1. Speakers

Speech material was taken from the TEVOID Corpus (Dellwo et al., 2012; Leemann et al., 2014). For this database 16 speakers (8m, 8f) of Zurich German were recorded (mean age=27 years, SD=3.6, range: 20–33). Most of the speakers were students at the University of Zurich and were screened for their regional vernacular (Zurich German; the urban variety of the vernacular) prior to the recordings by trained phoneticians (second and third authors). All speakers were found to be free from any speech or hearing disorders. Recordings took place in a sound-treated booth in the Phonetics Laboratory at the University of Zurich using a Neumann STH-100 transducer microphone (sampling rate=44.1 kHz, quantization level=16 bit). Each speaker was paid 30 Swiss Francs per hour for the recording session and the time for the preparation of reading the sentences. 4 female (1, 4, 5, 6) and 4 male speakers (7, 12, 14, 15) were selected randomly from the TEVOID corpus for the present experiment.

3.1.2. Listeners

26 listeners of Swiss German (6m, 20f) took part in the experiment. Listeners were all students at the University of Zurich and took part in the experiment voluntarily. They were paid 15 Swiss Francs for their participation, which took about 30 min.

3.1.3. Material

Stimuli were created from the speech material in the TEVOID Corpus (see Section 3.1.1). The material in this database consists of spontaneous and read speech.

3.1.3.1. Spontaneous speech. To collect spontaneous speech data, an interview on topics of general nature (e.g. holidays, fields of study, future professional plans) was carried out with each speaker. Each interview lasted around 45 min. The second and third authors carried out the interviews in Swiss German. From each interview, 16 sentences were extracted resulting in 256 sentences in

total (16 speakers*16 sentences). We looked for sentences of about 15 syllables in length, even though this number sometimes varied considerably. The obtained mean number of syllables was 12.8 (std dev=4.7) in the final dataset. Of all possible sentence candidates 16 sentences were randomly selected for each speaker. All 256 sentences (16 sentences*16 speakers) were transcribed lexically in Zurich German. The criteria for the utterance extraction from the spontaneous interviews were that the utterances were grammatically and semantically coherent and fluently produced. This was defined in the following way:

- *Grammatical coherence*: the utterances had to be free from grammatical mistakes or word repetitions. The lexical items of the utterances had to be meaningful items that occur in everyday speech.
- *Semantic coherence*: Utterances had to form meaningful units in themselves even when the interpretation of the utterance within the context of the interview might have been different.
- *Fluent reading*: Sentences had to be free from interruptions, hesitations, filled or unfilled pauses.

3.1.3.2. Read speech. A few months after the interviews, the 16 speakers were re-invited to read the transcribed sentences resulting in a total of 4096 read sentences (16 speakers*256 sentences). As there is no standard writing system for Swiss German, we created transcriptions particularly tailored for Zurich German. This was done based on the speakers' expected experience with written Swiss German. Given that our speakers were all university students, we felt it most appropriate to opt for a balance between writing the spontaneous sentences as phonetically as possible while at the same time following a spelling close to Standard German. As reading skills of vernacular transcripts varied between the speakers they were provided with the transcript a few days prior to the recordings and were asked to practice reading them. All speakers were able to read the sentence list fluently at the time of recording. Examples of the sentence transcripts are provided in [Appendix 1](#).

3.1.3.3. Stimulus set. The stimulus set was created by random selection of 4m and 4f speakers from the TEVOID database of which 8 spontaneous and 8 read sentences were, again randomly, selected. This made up a total of 128 stimuli (8 speakers*(8 spontaneous sentences+8 read sentences)). All stimuli were normalized for intensity by setting the average sound pressure to 70 dB SPL. For the read sentences each sentence occurred only once. The number of syllables per sentence for our subset was representative for the figures obtained from the entire database (see above): mean=12.64, std dev=4.45. The figures are also represented by the two speaking style subsets of the database (read: mean=12.89, std dev=4.09; spontaneous: mean=12.39 std dev=4.81). In addition we compared the mean number of syllables per sentence between speakers and found that they were very similar (numbers in brackets contain the standard deviation): speaker 1: 11.13 (3.36); speaker 2: 11.38 (4.24); speaker 3: 13.25 (4.36); speaker 4: 13.19 (3.51); speaker 5: 13.56 (6.86); speaker 6: 13.31 (3.84); speaker 7: 12.81 (4.09); speaker 8: 12.5 (4.76). Only speaker 5 had a higher standard deviation of 6.86 which resulted from a single sentence containing 36 syllables. No effects of the number of syllables in a stimulus could be obtained (see below).

3.1.4. Procedure

Listeners were tested on a laptop computer with high-quality closed headphones (Beyerdynamic, DT 770 PRO; 250 Ω) in a quiet environment at the Phonetics Laboratory of Zurich University. The demo window function along with tailor-made scripts in Praat ([Boersma & Weenink, 2014](#)) was used for stimulus presentation. Listeners were presented with a screen on which the question *War das vorgelesen oder frei gesprochen?* (literally: 'Was this read or freely spoken?') would be typed out in red letters in the upper half of the screen. Below the question there were two large buttons next to each other, the left one was labeled *vorgelesen* ('read') and the right one *frei gesprochen* (literally: 'freely spoken'). Both experiments, I and II, were performed with the same listeners in one session taking approximately 30 min altogether. Each listener was tested individually. For Experiment I listeners were instructed that they had to rate utterances as to whether they were spontaneously produced or read. To illustrate what spontaneous and read means, listeners were told to imagine that they listened to a person speaking on the radio and that it would typically be easy to decide whether someone reads a text (e.g. news) or whether they produce speech spontaneously (e.g. a debate). Before the experiment started, listeners received a short demonstration of the interface during which they responded to two example stimuli (one typical read and one typical spontaneous stimulus, as judged by the authors) that were not part of the experiment.

3.1.5. Statistical analysis

The bias free non-parametric sensitivity measure A' from Signal Detection Theory was used to measure listeners' identification performance ([Snodgrass & Corwin, 1988](#)). Read and spontaneous conditions were arbitrarily attributed to be target and noise, i.e. read speech presented and read speech responded was a *hit*, spontaneous speech responded to spontaneous speech was a *correct rejection*. Equally the two error types consisted of the following: read speech presented and spontaneous speech responded was a *false negative* and spontaneous speech presented and read speech responded a *false positive*. The A' scale ranges from 0 to 1, with 0.5 being chance level (no sensitivity). For the present situation this would mean that in case of an A' value of 0 a listener would have systematically accepted read speech as spontaneously produced and vice versa. An A' value of 1 would mean that a listener systematically responded to all stimuli correctly. To investigate listeners' bias we calculated B_D' which extends from -1 to 1. A B_D'

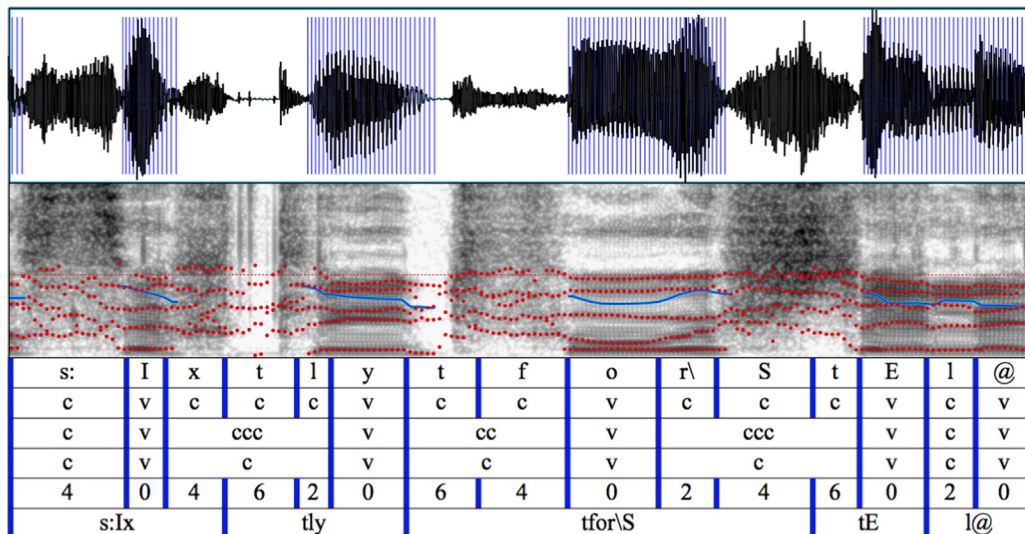


Fig. 1. Example of the beginning of an utterance from the TEVOID corpus with six interval tiers containing segmental information (1), consonantal and vocalic interval information (2, 3 and 4), sonority information for each segment (5) and syllabic durational information (6).

value of 0 indicates no bias, a positive value indicates a bias towards spontaneous speech and a negative value indicates a bias towards read speech (Pallier, 2002). All statistical analyses were carried out with R (R Core Team, 2013).

3.1.6. Measures of acoustic variability of read and spontaneous speech

Eight measures have been used to measure the acoustic variability between read and spontaneous speech. All measurements were performed in Praat using scripts written by the first author. Measurements were based on temporal segmental information available for all files in the TEVOID corpus as can be seen in Fig. 1. The first tier in the figure containing segmental durational information was manually attributed to the data by the second and third authors of this paper. The following tiers were attributed semi-automatically. Tier 2 contains the corresponding labels of whether a segment is a consonant or a vowel. Tiers 3 and 4 contain the consonantal and vocalic interval information (Grabe & Low, 2002; Ramus et al., 1999) with and without the number of underlying segments. Tier 5 contains sonority levels for each segment on the basis of which the syllable boundaries in tier 6 were obtained automatically (cf. Leemann et al., 2014, for the method).

- *Articulation rate*: The number of syllables per second excluding all silent parts in the speech signal longer than 50 ms measured in Tier 6 of the TextGrid (Fig. 1).
- *F0 average*: A fundamental frequency contour (Pitch object) was extracted using Praat's "To Pitch..." function for each stimulus. To prevent measuring mistakes we adjusted the frequency range for each gender (male: 90–250 Hz, female: 150–300). Each pitch contour was plotted and checked by visual inspection by the first author and in the event of major errors (e.g. octave jumps) the contour was corrected. The mean fundamental frequency for each stimulus was obtained with Praat's function "Get mean..." in the Pitch object.
- *F0 variability*: From the same Pitch object the standard deviation of F0 in each stimulus was obtained with Praat's function "Get standard deviation...". Between-speaker F0 variability (in particular male–female) was normalized for by calculating the coefficient of variability of F0 ((F0 standard deviation*100)/mean F0).
- *Intensity variability*: An intensity object was created for each stimulus in Praat. Based on the vocalic interval information in tier 4 (Fig. 1) the mean intensity was measured for each vocalic interval in the intensity object. The coefficient of variation was calculated for all obtained intensity mean values in each stimulus ((intensity standard deviation*100)/mean intensity).
- *Long-term spectral slope (LTAS)*: A pitch-corrected long-term average spectrum (LTAS) was created for each stimulus (Praat function: To Ltas (pitch-corrected)...). This function only considers voiced intervals of the signal and normalizes for F0 influences.
- *Vocalic-consonantal durational ratio (%V)*: This was calculated as the percentage over which speech is vocalic (%V, Ramus et al., 1999) based on the consonantal and vocalic interval durational information in tier 4 (Fig. 1).
- *Consonantal durational variability (deltaCln)*: The standard deviation of consonantal variability was calculated based on tier 4 (deltaC, Ramus et al., 1999). To correct for skewness of consonantal interval duration distributions and to normalize for speech rate we applied the method suggested by Dellwo (2009) by which deltaC is calculated based on logarithmically-transformed consonantal interval durations (deltaCln).
- *Vocalic durational variability (nPVI_V)*: The rate-normalized Pairwise Variability Index of vocalic intervals was calculated (nPVI_V, Grabe & Low, 2002) based on the vocalic interval duration information on tier 4 (Fig. 1). For this measure the average of all differences between consecutive vocalic intervals were calculated. To normalize for speech rate influences, the interval difference was calculated proportionally to the total duration of the respective interval pair (nPVI_V).

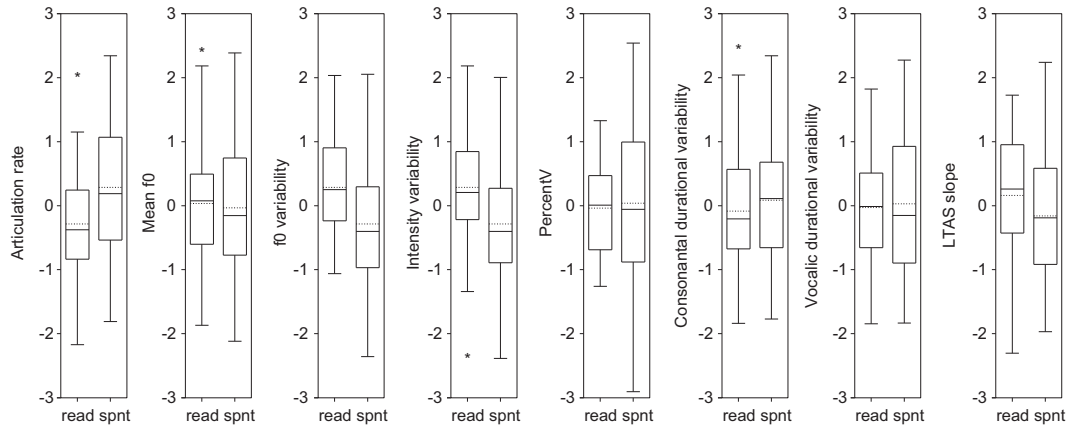


Fig. 2. Box-plots of the distributions of eight acoustic variables for each read (read) and spontaneous (spnt) speech. Values were z-score normalized by speaker to remove speaker-specific variability. Dotted lines in the boxes indicate the mean, solid lines the median.

Table 1

F-values for a two-way ANOVA for the eight acoustic variables under investigation (column 1). Main effects for speaking style (column 2) and speaker (column 3) and the interaction of the two main effects (column 4) are provided. Significant effects ($p < 0.01$) are flagged with single and highly significant effects ($p < 0.001$) with double asterisk.

Acoustic variable	Speaking style (df: 1)	Speaker (df: 7)	Interaction (df: 7)
Articulation rate	14.47**	8.22**	0.25
F0 Mean	3.86	296.93**	2.92*
F0 variability	15.09**	3.81**	2.54
Intensity variability	12.6**	6.61**	2.98*
%V	0.11	2	0.8
δCLn	0.32	1.54	0.95
$nPVI-V$	0.002	1.52	1.99
LTAS slope	36.02**	4.93**	22.23**

3.2. Results

3.2.1. Acoustic differences between read and spontaneous speech in our stimuli

Fig. 2 contains box-plots of the acoustic variables under investigation for read and spontaneous speech. Values have been z-score normalized by speaker to remove speaker-specific variability $z = (\text{value} - \text{mean for speaker}) / \text{standard deviation for speaker}$. The plots show that articulation rate tends to be higher in spontaneous speech in our data, and F0 and intensity variability as well as the LTAS slope tend to be lower. There appears to be no difference between read and spontaneous speech for the other parameters observed. It has to be noted too, however, that the within-category variability is strong and even though the mean differs notably between styles for some of the variables under investigation, the overlap between the distributions is high in any case.

Variability of the acoustic parameters between read and spontaneous speech was tested in a two-way ANOVA with a second factor speaker. The factor speaker was included to test for interactions between speaker and speaking style, i.e. to test whether individual speakers behave differently in their acoustic distinction of read and spontaneous speech (hence no z-score normalization for speaker was applied). Table 1 contains the results of the analysis. Because we carried out multiple tests on the same dataset we tested at a conservative alpha level of 0.01 (significance) and 0.001 (high significance). The results revealed that both articulation rate and F0 variability varied highly significantly between read and spontaneous speech and did not interact with likewise highly significant effects for speaker. This means that all speakers, even though they varied in their individual level of articulation rate and F0 variability, maintained the articulation rate differences between the read and spontaneous speech. For intensity variability and LTAS slope we also observed highly significant effects between speaking styles, however we also observed significant interactions (highly significant in the case of LTAS slope) with the likewise highly significant main effects for speaker. This means that individual speakers treat intensity variability and LTAS slope differently between the speaking styles. It is thus less likely that these two parameters serve listeners in the distinction of read and spontaneous speech.

The results of the speech rhythm variables (%V, δCLn and $nPVI-V$) replicated the results of our previous studies on read and spontaneous speech differences on the larger dataset where the stimuli have been taken from (Dellwo et al., 2012; Leemann et al., 2014). This confirms the finding that for our particular dataset there are no systematic differences in rhythmic characteristics between the read and spontaneous speech and thus lets us assume that listeners cannot make use of these variables to distinguish between the two different speaking styles.

The most likely variables that listeners can base their distinction on are articulation rate and F0 variability. Both show variability that is independent of the speaker producing it. Articulation rate is higher in the spontaneous data and F0 variability is lower. It is possible that intensity variability and LTAS slope support listeners' identification ability of read and spontaneous speech, however, because of the interaction with speaker this seems rather unlikely.

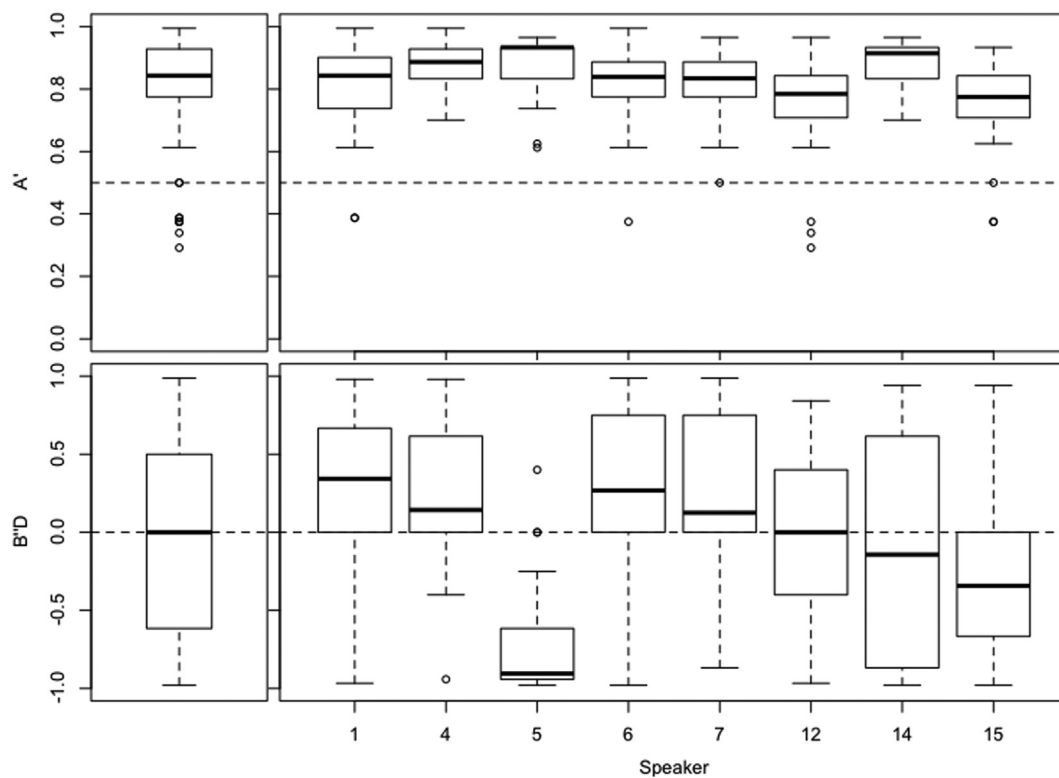


Fig. 3. Distributions of A' (top) and $B'D$ (bottom) for all speakers (left) and subdivided by speaker (right); $N=208$ (26 listeners*8 speakers).

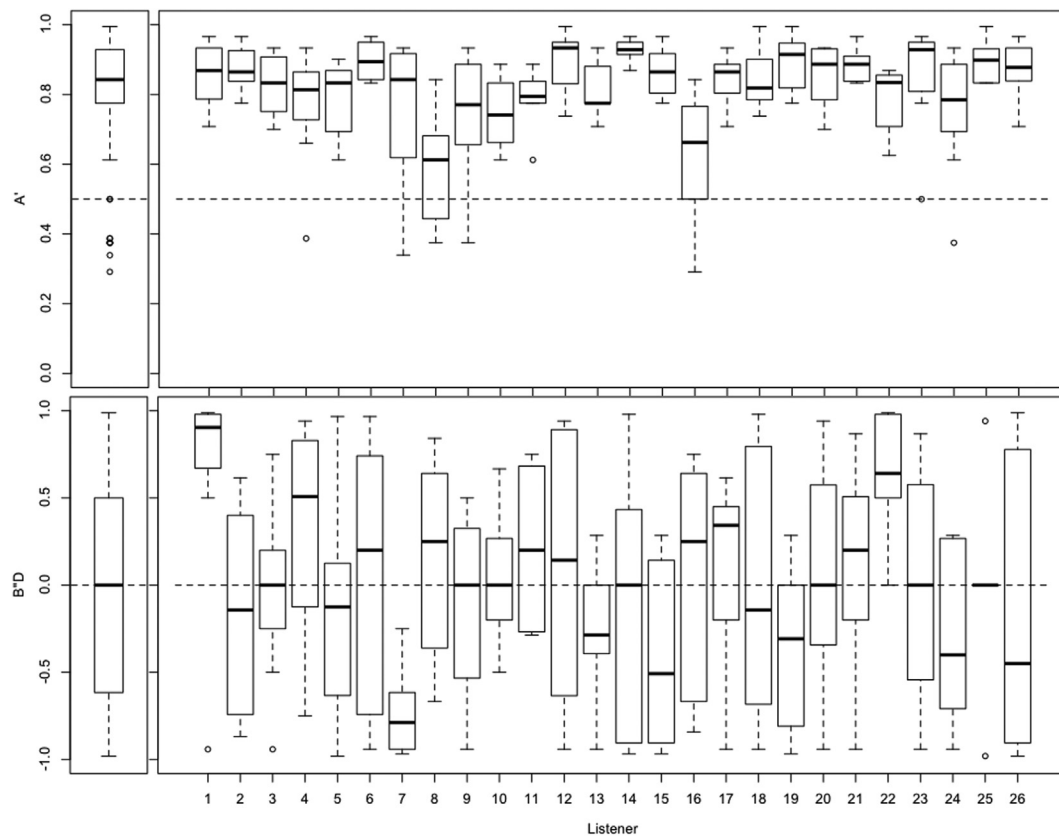


Fig. 4. Distributions of A' (top) and $B'D$ (bottom) for all listeners (left) and subdivided by listener (right); $N=208$ (26 listeners*8 speakers).

Table 2

R-squared values for a linear regression model between the acoustic variable (column 1) and listener accuracy for all data (column 2), read data only (column 3) and spontaneous data only (column 4). Significant values are flagged with (*; $\alpha=0.0003$), highly significant values with (**; $\alpha=0.003$).

Dependent	<i>r</i> -Squared (all data)	<i>r</i> -Squared (read)	<i>r</i> -Squared (spontaneous)
Articulation rate	0.02	0.09	0.25**
F0 mean	0.02	0.01	0.03*
F0 variability	0.08**	0.08	0.09
Intensity variability	<0.01	0.04	0.03
%V	0.02	0.02	0.02
<i>deltaCLn</i>	<0.01	0.05	0.03
<i>nPVI-V</i>	<0.01	0.03	0.01
LTAS slope	<0.01	0.01	<0.01

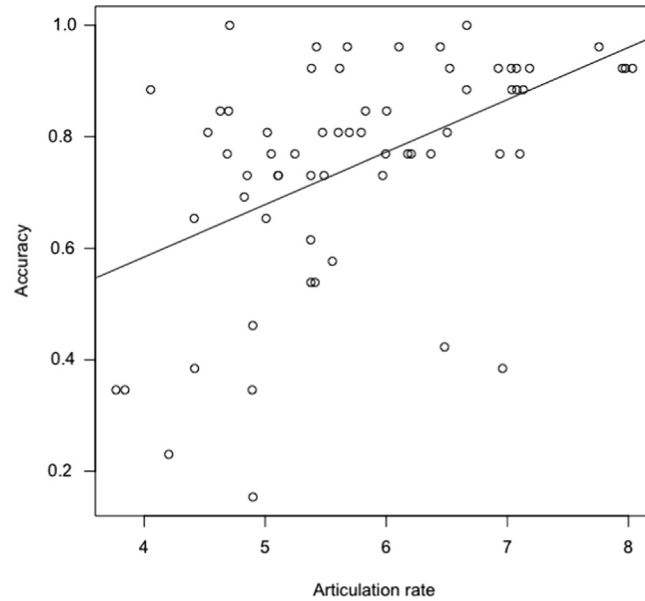


Fig. 5. Scatter plot of listener accuracy in Experiment I as a function of articulation rate for the spontaneous data with a superimposed regression line.

3.2.2. Perceptual results

A' and B'_D were calculated for each listener for each speaker making for $N=208$ (8 speakers*26 listeners). The listeners' overall performance was very high (Fig. 3, top). Fig. 3 (top, left) reveals that the central 50% of the listener data had A' scores between 0.8 and 0.9 with the upper range of the distribution approximating maximum sensitivity of 1.0. This effect was highly significant (testing the difference from chance level (i.e. $A'=0.5$) in a one-sample t -test: $t(207)=86.6$, $p<0.005$). Listeners were thus able to solve the task to a very high degree. Within this high performance, however, we found between-speaker variability (Fig. 3, top, right). An ANOVA of A' by speaker with repeated measures on listener and sentence (R code: `aov(APrime~speaker+Error(listener/sentence))`) revealed that this effect was highly significant ($F(7,149)=6.62$; $p<0.001$). To test whether listeners' performance was above chance for each speaker, a one-sample t -test was performed for each speaker (Bonferroni adjusted alpha: 0.006 (0.05/8) for significant and 0.0006 (alpha/10) for highly significant effects); $t(27)$ ranged between 7.04 and 27.78; each test was highly significant.

Listeners' overall bias was spread widely, ranging from -1 to 1 (Fig. 3, bottom, left). There was strong between-speaker variability of bias, however. While for speakers 1, 4, 6 and 7 listeners' bias was positive (i.e. these speakers were predominantly perceived spontaneously-speaking), bias was strongly negative for speaker 5 and negative for 15 (i.e. speakers were predominantly perceived as read). Bias for speakers 12 and 14 was spread rather evenly between -1 and 1 . An ANOVA of B'_D by speaker with repeated measures on listener and sentence revealed a highly significant effect for between-speaker bias differences ($F(7,149)=12.36$; $p<0.001$).

Performance varied between individual listeners (Fig. 4, top, right); the effect was significant (ANOVA: A' by listener: $F(25,182)=3.97$; $p<0.001$). However, the performance differences are typically weak in magnitude and, apart from very few exceptions (listeners 8 and 16, for example), vary on a high performance level in the range of about 0.7 and 0.9. To test whether individual listeners were significantly above chance, we carried out a one-sample t -test for each listener (Bonferroni adjusted alpha: 0.002 (0.05/26) for significant and 0.0002 (alpha/10) for highly significant effects); $t(7)$ ranged between 9.02 and 82.45. For 6 listeners the effect was not significant (4, 7, 8, 9, 16 and 24), two listeners performed significantly above chance (10, 23), the remaining 18 listeners performed highly significantly above chance.

When it comes to listeners' bias we found that listeners range over the full bias scale from -1 to 1 and that listeners vary to a considerable degree between each other. Some listeners have a strong positive bias (1 and 22), other listeners have a strong negative bias (7 and 19). For many listeners, however, bias varies between positive and negative values (e.g. 6, 12 or 26). Only one listener seems virtually bias-free (25) apart from one positive outlier.

3.2.3. How well do the acoustic stimuli explain listener rating behavior?

Given the variability of some acoustic variables, in particular articulation rate and F_0 variability (Section 3.2.1), we tested the degree to which listener identification ability could be explained by the acoustic variability between read and spontaneous speech. To do this, we calculated the accuracy for each stimulus (average between hit rate for read and hit rate for spontaneous responses). Accuracy ranges between 0 (all responses incorrect) and 1 (all responses correct).

To calculate how well listener accuracy could be predicted by each of the acoustic variables under investigation we fitted a linear regression model to each variable pair (R-code: `lm(accuracy~[acoustic variable x])`). The r -squared values are presented in Table 2. An f -test was carried out to test the significance of r -squared. Since we split our dataset into three parts we Bonferroni-corrected our conservative alpha levels of 0.01 by dividing alpha by three (significance: 0.003, high significance: 0.0003). Table 2 reveals that accuracy can only be predicted very poorly by the prosodic variables under investigation. The only two cases in which high significance could be reached was F_0 variability for all data and articulation rate in case of the spontaneous data. In case of F_0 variability the prediction of accuracy is miniscule (8% of the variance of accuracy can be predicted by F_0 variability) and in case of articulation rate it is low but clearly present (25% of the accuracy variance can be predicted).

Scatter plots of the acoustic variable under observation and listener accuracy were studied visually to ensure that the linear models reflected the data situation well. It was found that there was no reason to assume alternative relationships between the variables. For the accuracy* articulation rate pair this scatter plot is provided in Fig. 5. As the regression line shows a positive inclination, the relationship is based on a positive covariance between the variables, i.e. higher rates result in a higher accuracy for the spontaneous data.

To test whether there is evidence for the assumption that all our acoustic variables in combination might be a better predictor for listener accuracy, we fitted a model with all of them (R code: `lm(accuracy~sum of all acoustic variables)`). The adjusted r -squared value of this model was 0.09. This means that the combination of variables did result in a stronger prediction than individual variables. The same model applied for the spontaneous data returned an r -squared of 0.23, which is even smaller than the r -squared reached with articulation rate alone for this data. The model for the read data returned an adjusted r -squared value of 0.15, which is slightly higher than the prediction power of any of the variables on their own but still very small.

In summary, in the acoustic data analysis above we found evidence that of all acoustic variables under investigation, only articulation rate and intensity variability differ between read and spontaneous speech in our stimulus set. For LTAS and F_0 variability we also found significant effects, however, the interaction with speaker suggests that the effects can vary between speakers and are thus difficult to generalize. Fitting linear regression models to test how much of listener accuracy for individual stimuli can be explained by the acoustic variables, we found that these numbers were very low and in most cases not statistically significant. We therefore conclude that listeners did not extensively use the acoustic prosodic variables under investigation to distinguish between read and spontaneous speech in our experiment.

We also calculated regression coefficients between listener performance and each of the eight acoustic variables under investigation for each of the eight speakers. Since we split the dataset in eight subsets (one for each speaker) we lowered our alpha level to 0.0012 (0.01/8). Even though we occasionally obtained higher r -squared values for some of the variables (between 0.4 and 0.6), the p -values for none of these effects came close to our alpha threshold. We must therefore conclude that attention to the acoustic variables does not vary between individual speakers.

Finally, we reported varying number of syllables between stimuli (Section 3.1). To ensure that listener performance was not influenced by this number we also ran our linear regression models on the accuracy * number of syllables variable pair. All r -squared regression coefficients were below 0.07 and none of the effects was significant. It must be concluded that the number of syllables in stimuli did not influence listeners' accuracy.

3.3. Discussion

Results from the present experiment could not be clearer: Our null hypothesis that the read and spontaneous speech in our TEVOID corpus cannot be reliably identified must be rejected. On the contrary, with an A' interquartile range between 0.8 and 0.95 we can conclude that listeners' performance is very high and the very large majority of listeners (about 80%) were well able to perform the task. High identification performance was found for all speakers in our database. Although A' revealed a significant between-speaker effect, these differences were found to be of small magnitude. Read speech of all speakers could be identified reliably and to a high degree. There is no reason to assume that the differences between read and spontaneous speech are not well realized by individual speakers in our database. The two speaking styles could be identified well despite a lack of typical spontaneous cues and despite the ambiguous cues to read speech (Swiss vernacular productions; see Section 1).

There was some evidence in the data for our assumed bias of listeners towards a preference of identifying the stimuli as spontaneous speech. This bias, however, was strongly dependent on particular speakers; four out of eight speakers resulted in a strong spontaneous listener bias. In addition, listener bias varied between individual listeners. While about seven listeners showed a bias towards spontaneous speech, about the same number of listeners revealed an opposite bias. It must therefore be concluded

that our assumption about a listener bias is only true for some listeners. We assume that individual usage preferences of Swiss German might play a strong role in the formation of such listener biases. Listeners that are less confronted with read vernacular speech in everyday situations might show a stronger spontaneous bias. We currently have no explanation for why some listeners were stronger biased towards read speech in our experiment.

What are the acoustic cues for the high listener performance? The analysis of predominantly prosodic cues did not reveal clear indicators. Even though we obtained significant differences in articulation rate, F0 and intensity variability as well as LTAS slope characteristics it seems that our listeners hardly, if at all, paid attention to these cues in their decision process. The only notable indicator was articulation rate in the spontaneous condition, which can predict around 25% of listener response variance with a positively inclined regression line. This data is in line with findings by Laan (1997) who reported higher articulation rates for spontaneous speech in two speakers of her study. In our data we were able to show that articulation rate contributes to the accuracy of listeners to some degree when identifying spontaneous utterances. Numerous other studies reported lower articulation rates for spontaneous speech (Hirschberg, 2000; Levin et al., 1982), however, in these cases measures of articulation rate also included hesitations and pausing characteristics. Our own and Laan's results provide evidence that the filled speech parts in spontaneous speech are cues to listeners' identification of the speech as spontaneous.

4. Experiment II

Here we tested the degree to which listeners perceived read speech as read and whether this degree varied between speakers and according to sentence origin. For each speaker, half of the sentences they read derived from spontaneous utterances they previously produced themselves and the other half derived from spontaneous utterances produced by their peers. We further tested whether the acoustic prosodic variables introduced in Experiment I can explain listener rating choices.

4.1. Method

4.1.1. Listeners

The same 26 listeners from Experiment I also participated in Experiment II.

4.1.2. Stimuli

The 8 read sentences from each of the 8 speakers (total 64 stimuli) from Experiment I served as stimuli for the present experiment (see Section 1 for the rationale for using read instead of spontaneous speech). To test the influence of sentence origin (sentences originated from the speakers' own or other speakers' spontaneous speech), 4 of the read sentences for each speaker were taken from the set that s/he previously produced spontaneously and 4 sentences were randomly taken from other speakers' previous spontaneous productions.

4.1.3. Acoustic variables

The acoustic variables described in Experiment I were applied.

4.1.4. Procedure

Praat (demo window function) was used for stimulus presentation (see Section 3.1.4). Listeners were presented with a screen on which the question *Tönt das eher mehr oder weniger vorgelesen?* (literally: 'Does this sound rather more or less read?') was typed out in black letters in the upper half of the screen. In the center of the screen there was a gray continuous scale (subdivided into 100 equidistant intervals that were not visible to the participants) reaching from *ehrer weniger vorgelesen* (literally: 'rather less read') on the left end to *ehrer mehr vorgelesen* (literally: 'rather more read') on the right end of the scale. Listeners were instructed to rate utterances as to whether they appeared more or less read to them. They were told that they should indicate their impression on the scale described above and they were encouraged to use the entire scale. Since each participant carried out Experiment I right before Experiment II, listeners knew about the concept of read and spontaneous speech (Section 3.1.4). To avoid listeners from assuming that some of the stimuli might be spontaneously produced they were told that all sentences they would hear were read speech but that there are cases where this read speech might appear more or less like read speech. To familiarize participants with the experiment interface, they received a short demonstration of two randomly selected read sentences from the TEVOID corpus (not part of the experiment) prior to the start of the experiment.

4.2. Results

4.2.1. Perception results

Rating results for each speaker and sentence origin are plotted in Fig. 6. It can be seen that there is strong variability between the speakers, where speaker 5 was strongly perceived as read. We obtained the most visible difference in sentence origin for speaker 12 where the reading of his own sentences was rated as much less read than those sentences of a different origin. A 8*2 factorial ANOVA (speaker*sentence origin) revealed highly significant main effects for both factors (speaker: $F(7,400)=52$; $p<0.001$; origin: $F(1,400)=7.86$; $p<0.001$) but there was also a highly significant interaction ($F(7,400)=8.68$; $p<0.001$) which makes the main effects not readily interpretable. For this reason simple effects were calculated for speaker and sentence origin. For the simple effects of speaker (df: 7, 200) we used a Bonferroni adjusted alpha: 0.025 (0.05/2). We obtained highly significant effects for both sentence

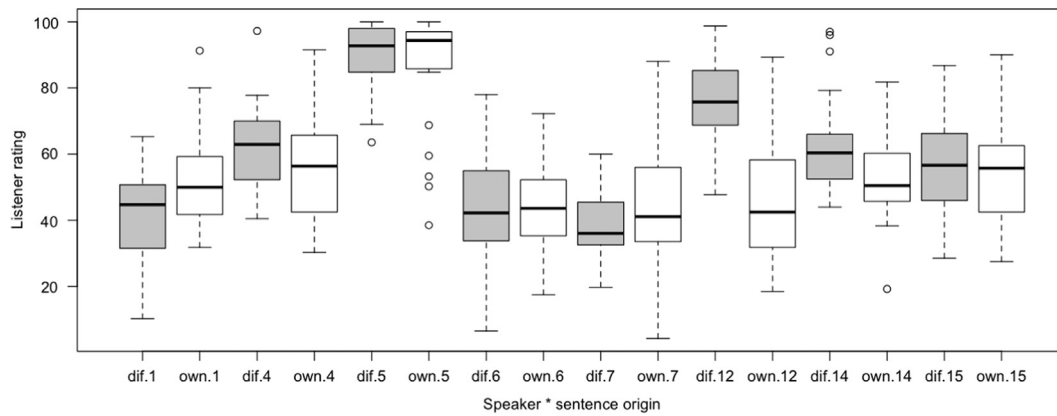


Fig. 6. Distributions of listener ratings (averaged by listener) as a function of 8 speakers subdivided by sentence origin (gray=sentences derived from different speakers; white=sentences derived from the speaker herself/himself); $N=416$ (26 listeners*8 speakers*2 origins).

Table 3

R -squared values for a linear regression model between the acoustic variable (column 1) and listener ratings.

Acoustic variable	r -Squared
Articulation rate	0.44**
F0 mean	0.05
F0 variability	0.03
Intensity variability	0.03
%V	0.09
δCLn	0.05
$n\text{PVI-V}$	0.03
LTAS slope	<0.01

origins (own: $F(7, 200)=19.5$; $p<0.0025$; different: $F(7,200)=45.3$; $p<0.0025$). In the simple effects of sentence origin (df: 1, 50) we applied a Bonferroni adjusted alpha: 0.006 (0.05/8). The effect was significant in case of speaker 1 ($F(1, 50)=8.1$; $p=0.006$) and highly significant for speaker 12 ($F(1, 50)=44.5$; $p<0.0006$). We did not find significant effects for any of the other speakers.

When we compare the *own* and the *different* plots between speakers separately we find that there is much more overlap between the central two quartiles in the *own* condition (apart from speaker 5) than in the *different* condition. To quantify this effect, between-speaker differences were measured post-hoc with a Bonferroni adjusted pairwise t -test (R code: `pairwise.t.test(response, speaker [own/dif], p.adj="bonferroni")`). With 8 speakers, 28 pairwise comparisons were possible. In the "own" condition, 25% (7/28) of the pair-comparisons were significant and in the "different" condition, 79% were significant (22/28). This means that the speakers' reading was judged much more variably when they produced sentences that originally stemmed from another speaker.

4.2.2. Listener rating behavior as a function of the acoustic variables

Like in Experiment I, we fitted linear regression models to all variable pairs resulting from pairing each of the acoustic variables with listener rating (R-code: `lm(rating~[acoustic variable x])`) and tested the significance of the r -squared values with an f -test (identical alpha values like in Section 3.2.2). Table 3 contains the r -squared values of this linear regression analysis.

In the present case articulation rate is able to predict a more notable amount of listener data, i.e. 44%. Fig. 7 contains a scatter-plot of listener rating as a function of articulation rate. We found in the plot that the values of one speaker (5) were predominantly scattered in the upper left corner of the plot and thus we assumed that the result was influenced by this particular speaker. Speaker 5 reveals some of the lowest syllable rates and received the highest reading ratings. For this reason we fitted the same regression model on a dataset in which we excluded speaker 5. The resulting r -squared of 0.25 was still highly significant but in this model articulation rate thus only explained 25% of the variance of the listener rating.

Finally, we also tested whether the varying number of syllables between stimuli might have had an influence on listeners' rating behavior and fitted the linear regression model to this variable pair. The obtained r -squared was 0.04 and the effect was insignificant. In this second experiment, too, there is no evidence that the total number of syllables per sentence had an effect on listeners' response.

4.2.3. Listeners' rating behavior as a function of their accuracy

To test whether the listeners' accuracy for the read stimuli from Experiment I is related to the listeners' rating behavior in Experiment II we fitted a linear regression model to the two variables (R-code: `lm(accuracy~rating)`). In other words, we tested whether we can predict listeners' rating behavior (whether a stimulus sounds more or less read) by their accuracy about a stimulus (is a read stimulus identified as read or not). The r -squared for accuracy by rating is 0.54 ($p<0.001$). This means that 54% of the rating behavior variance in Experiment II can be predicted from the listeners' accuracy variance in the forced choice task in Experiment I.

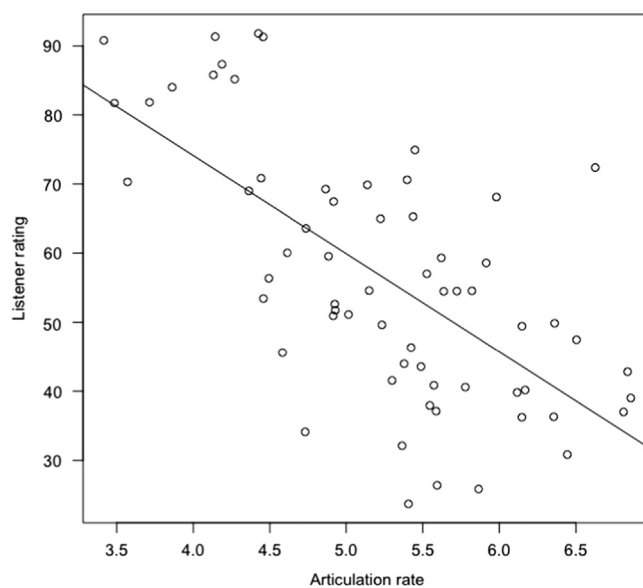


Fig. 7. Listener rating plotted across articulation rate for the 64 read stimuli in Experiment II.

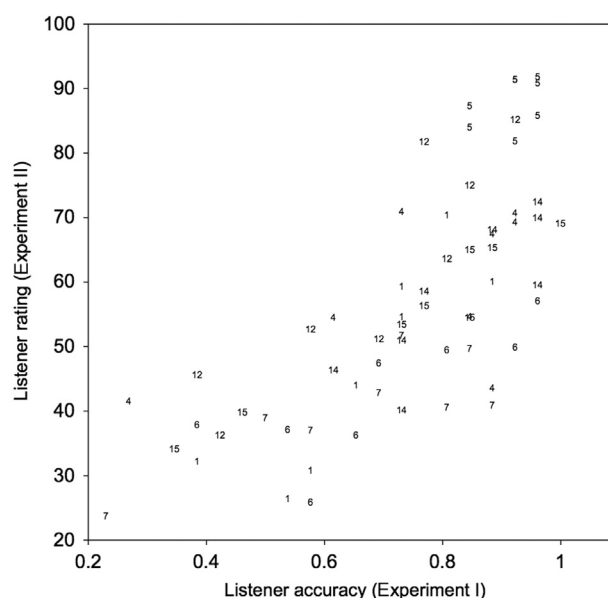


Fig. 8. Listener rating from Experiment II (y-axis) plotted across listener accuracy from Experiment I (x-axis) for the 64 read stimuli.

Fig. 8 reveals again that the sentences of speaker 5 mark the top right of the distribution and might thus have influenced the regression result strongly. A model without speaker 5, however, resulted in roughly the same r -squared value (0.52; $p < 0.001$).

4.3. Discussion

Given the results from Experiment II, it appears that the read speech of some speakers is perceived as more representative in terms of read speech than that of other speakers. What are the reasons for this variability? Again, we found articulation rate to be one of the indicators of read speech ratings. This time, around 45% of the variance of listener rating could be explained by articulation rate. Even though this figure was to some degree influenced by one particular speaker, there is evidence that listeners used articulation rate as an indicator to read speech characteristics. The slope of the regression line is negative indicating that faster speech was rated as less read than slower speech. This result is in line with the one from Experiment I where we found a positive relationship between articulation rate and obtained listener accuracy in spontaneous speech. So higher articulation rates lead to a lower rating of read speech as read (Experiment II) and to a higher accuracy of spontaneous speech to be identified as spontaneous (Experiment I). The question that remains is why listeners did not use the articulation rate cues in the identification task. In other words, why did we obtain such a low regression coefficient between listener accuracy and articulation rate in Experiment I? On the one hand it might be that our listeners involved different strategies in the two different tasks. It seems possible that there were enough alternative cues in the identification task that indicated read speech as read and listeners paid less attention to articulation rate cues.

In Experiment II, when listeners were only presented with read speech, it seems that rate differences became more important. This situation might be evidence for the hypothesis that acoustic cues to read and spontaneous speech are probably not static and are not interpreted equally under any given circumstance but that cues are used in variable ways according to their availability. It thus seems plausible to assume that there might be a hierarchy in cues to read and spontaneous speech. The results from our experiments suggest that articulation rate is not ranked the highest but is applied when other important cues are missing.

Given the variable performance as a function of speaker there may be alternative explanations for the rating results we obtained that possibly interact in a complex fashion with global variables such as articulation rate. Even though all speakers had fluent reading skills, there were differences between the speakers in this domain and phonetic expert listeners of Swiss German (second and third authors) judge the reading skills of speaker 5 as least fluent of all of them. In addition, results from Experiment I showed that listeners' read and spontaneous speech identification sensitivity was highest for this speaker. Speakers 6 and 7, however, vary in their reading skills. While speaker 6 was judged as strong by the phonetic experts, this was not the case for speaker 7. This is exemplary evidence for an interpretation that, in particular with our database, good reading skills lead to speech that is perceptually less characteristic in terms of read qualities. Such a result seems not all too surprising given the fact that the sentences that speakers are reading originally derived from spontaneous speech. If maximum differences between read and spontaneous speech in the laboratory should be desirable it might thus not be advantageous to use speakers that have high quality reading skills to the extent that they can imitate the spontaneous speech closely. Yet it might be equally undesirable to recruit speakers with poor reading skills as this might lead to some random variability, in particular concerning prosodic characteristics.

Sentence origin had a significant effect on the perception of read speech only for two speakers (1 and 12) and the effect was in opposite directions. In the case of speaker 12 the magnitude of the difference was rather large. The speaker was perceived as more read when producing sentences by other speakers compared to his own. We take this as evidence that this speaker reproduced his own sentences in a different way compared to the sentences of other speakers. The auditory differences were present by a much less fluent production of sentences stemming from other speakers (as judged by the phonetic experts). As hypothesised in the introduction, it is possible that the personal lexical and grammatical choices of the speaker may facilitate him/her to read these sentences more fluently. Our example thus provides evidence for the view that sentence origin may have an effect on the production of speakers' speaking style in read speech.

With speaker 1, we obtained an effect in the opposite direction. An explanation for this might be the sentence sequencing in the reading task. Sentences were read in the same order by all speakers from the transcript. Coincidentally, the transcribed sentences that originated from the spontaneous speech of speaker 1 occurred right at the beginning of the transcript, at the beginning of the recording session. Even though we gave speakers the chance to read through the sentences again, shortly before the recording, they might have felt slightly less relaxed when the actual recording started. Such effects indicate that if one intends to keep between-speaker variability to a minimum in laboratory recordings, it is advisable to make listeners more acquainted to the recording environment under actual recording conditions before having them read the experimental speech material.

Interestingly, a larger number of between-speaker differences could be found post-hoc for sentences derived from different speakers than for sentences derived from the same speaker, so speakers were more similarly rated when they reproduced their own sentences than when they reproduced sentences from other speakers. For the elicitation of read and spontaneous speech in the laboratory, this may mean that if we want to minimize between-speaker variability, it is of advantage to let speakers read their own sentences only, as has been done previously by [Laan \(1997\)](#) and [Mixdorf and Pfitzinger \(2005\)](#). More auditory variability between speakers might be obtained when speakers read transcripts of spontaneous utterances of their peers, even after prior practice, as in our method.

5. General discussion and conclusions

In the present study we tested listeners' perception of read and spontaneous speech derived from our TEVOID Corpus in two experiments. In Experiment I, listeners showed that they had a strong sensitivity to identify speech as either read or spontaneous. Experiment II revealed that, for the category of read speech, there can be strong differences between speakers according to how strongly listeners perceive their read speech as read. More between-speaker differences were observed under the condition in which speakers' read speech was previously produced spontaneously by other speakers (as opposed to produced by themselves). The only indicator to listeners' response behavior in the identification test (Experiment I) as well as in the rating (Experiment II) was articulation rate with a higher rate being a cue for spontaneity.

We introduced this paper by reporting on a specific question that came about from our previous research, namely that in our database spontaneous and read speech do not differ in speech rhythm (TEVOID Corpus; [Dellwo et al., 2012](#); [Leemann et al., 2014](#)). The results of the present study provide evidence for the view that the lack of these differences in temporal features is not due to a lack of audible differences between read and spontaneous speech. This means that speech rhythm, as measured in our research, does not seem to be responsible for the distinction of the audible read-spontaneous differences we obtained. This is also confirmed by the perceptual results of these experiments as we did not obtain any evidence for listeners basing their identification or rating performance on speech rhythmic characteristics. Given the results from [Remez et al. \(1986\)](#) this is surprising as they found, with a similar method, that after the removal of pitch cues, listeners were still able to discriminate between spontaneous and read speech. They argued that other prosodic cues, like speech timing, are more likely to be responsible for this effect. In future experiments it will be interesting to see whether this is true for our own data. It is possible that the relevant rhythmic/timing characteristics are not

obtained by the measures we performed (apart from articulation rate), but it is also possible that there are other, more segmentally related cues that mark the difference between spontaneous and read speech in this data.

Overall the experimental results are somewhat astonishing as the listeners showed high identification rates but the acoustic differences between the stimuli were not very large in magnitude (Fig. 2) and, apart from articulation rate, there was no evidence that listeners made use of any of the acoustic correlates of read and spontaneous speech to solve their tasks. Our effects for articulation rate, however, appear too low in magnitude to explain listener performance satisfactorily. The question thus remains what cues listeners used to achieve such high identification rates? Given that typical cues to spontaneous speech were missing in our stimuli (see Section 1) and that acoustically measurable prosodic variability is poor between styles, there are only segmental or voice quality cues left, which may serve as style indicators. Concerning the phoneme inventory, variability between read and spontaneous speech does not seem very plausible since there were no clear signs that the inventory changed between styles in the re-reading task. A possible and likely explanation is that our listeners were able to pay attention to much more fine grained cues than our acoustic prosodic analysis was able to reveal. It is also possible that such fine grained cues interact in a complex fashion, which was not part of our models. It will be interesting to look at this material in more detail in the future to be able to explain listeners' style perception ability better.

From both experiments we learned a number of points for the elicitation of read and spontaneous speech in the laboratory that we summarized in the following:

- *Overall method:* Our results show that laboratory elicited speech that lies at the borderline between read and spontaneous speech (see Section 1) contains sufficient cues to lead to a relatively high listener performance in identifying the two speaking styles.
- *Effects of listener:* Listeners varied significantly in their performance. A large number of listeners performed the task above chance. Our study provided support for previous findings that the performance of individuals generally plays a role in the identification of read and spontaneous speech (Laan, 1997; Mixdorff & Pfitzinger, 2005). The choice of listener thus seems vital in such recognition experiments, however, based on our data we are not in the position to characterize the listeners with low read-spontaneous speech recognition ability.
- *Effects of speaker:* Listener sensitivity varied significantly between speakers but performance was high for all speakers. The choice of speaker thus seems to play a minor role in eliciting read and spontaneous speech in the laboratory.
- *Within-category variability:* The read speech of some speakers was identified as more read than the read speech of others and this effect was significant. If between-speaker differences should be minimized, reading performance should be well controlled for when eliciting read speech.
- *Influence of sentence origin in read speech:* Between-speaker variability was higher when speakers reproduced sentences that had previously been spontaneously produced by other speakers, than when speakers reproduced sentences that they had previously produced spontaneously themselves. Again, if between-speaker variability should be minimized, it is desirable to use sentences that speakers previously produced themselves spontaneously.
- *Sequence of sentence presentation:* In one particular example (speaker 1) we found that an early occurrence of the speaker's own sentences is likely to have created an effect that we did not obtain elsewhere, namely that the speaker's own sentences were rated as more read than those of other speakers. To avoid such influences it is advisable to randomize the sentence order between subjects for the reading task, and possibly give readers a chance to rehearse in the environment of the recording booth prior to the recordings.

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Appendix I. Examples of stimulus sentences

The first 15 of the 128 transcribed utterances of the stimulus set:

- 1 So s Typische was sich d Lüüt vorschteled isch Kurator.
- 2 Ich bin wäge Spraachwüßeschaft dänn usegheit.
- 3 Das han i cool gfunde.
- 4 Mini Mueter isch ä no nie z Wien gsi.
- 5 Dänn mues ich ä no überlegge, was mis nöie Hauptfach wird.
- 6 Ich ha jetz äifach vergliichendi Spraachwüßeschafte gno.
- 7 Ich ha mich ä nie würlklich beworbe.
- 8 Chasch ja nöd nöime andersch go studiere mit Erasmus.
- 9 Zwar isch das Ganze im ne fiktive Königriich.

- 10 Ich wäis nöd werum si so abglänkt isch.
 11 Säge mer emaal ich fahr uf Oerlike.
 12 Mit em Zug sälber zwänzg Minute.
 13 Wil's äifach di zwäiti Spraach isch.
 14 Das git's äigentlich i käinere andere Spraach.
 15 Tschechisch han i käi Phonetik ghaa.

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